F22-SR-001
CARBON REGEN & ELUTION 415V SWITCHROOM

F22-SW-030
CARBON REGEN & ELUTION 415V MCC 1
2000-F22-TER-IC-10101

(NOTE 4)
F22-ETH-021
F22-ETH-023
INCOMER
F22SW030-GWY
F22SW030-INC INCOMER ACB
F22SW030-PQM POWER METER
F22SW030-AFD ARC FLT. DET. SYS.
RS485 BUS
EOL 1
EOL n
DOL STARTER MODULES ON FRONT & REAR OF MCC
(NOTE 1, 2 & 3)

LV VSDS
1
n

F22-DCS-412
CARBON REGEN & ELUTION 415V SWITCHROOM DCS PANEL
2000-F22-TER-IC-10011
F22-FCS-412
F22FCS412-NU01
(NOTE 1, 2 & 3)

F22-UPS-001
240VAC UPS

(NOTE 5)
F22-FIP-001
FIRE PANEL

F22-SW-031
CARBON REGEN & ELUTION 415V MCC 2
2000-F22-TER-IC-10121

(NOTE 4)
F22-ETH-024
F22-ETH-025
INCOMER
F22SW031-GWY
F22SW031-INC INCOMER ACB
F22SW031-PQM POWER METER
F22SW031-AFD ARC FLT. DET. SYS.
RS485 BUS
EOL 1
EOL n
DOL STARTER MODULES ON FRONT & REAR OF MCC
(NOTE 1, 2 & 3)

F22-COM-001
CARBON REGEN & ELUTION
415V SWITCHROOM COMMUNICATIONS RACK
2000-F22-TER-IC-10001

F22-ETH-001 (MODBUS COMMS SWITCH 1)
F22-ETH-002 (MODBUS COMMS SWITCH 2)
STACK CABLES NOTE 1.
LV SWITCHROOM MODBUS CONNECTIONS DISTRIBUTED ACROSS 2 ETHERNET SWITCHES

F22-ETH-011 (VNET IP1 LAN DCS SWITCH)
F22-ETH-012 (VNET IP2 LAN DCS SWITCH)

F22-FOB-001
F22-FOB-002
FIBRE BACKBONE THROUGH-PATCHING AS REQ'D
F22-ETH-003 (CCTV)
F22-FOB-003A
F22-FOB-003B
(NOTE 1, 2 & 3)

F22-SW-131
CARBON REGEN & ELUTION
EMERGENCY 415V MCC
2000-F22-TER-IC-10141

(NOTE 4)
F22-ETH-026
INCOMER
F22SW131-GWY
F22SW131-INC INCOMER ACB
F22SW131-PQM POWER METER
F22SW131-AFD ARC FLT. DET. SYS.
RS485 BUS
EOL 1
EOL n
DOL STARTER MODULES ON FRONT OF MCC
(NOTE 1, 2 & 3)

F22-KN-411 A-H
CARBON REGEN KILN ZONE 1
2000-F22-SCM-EL-10207
2000-F22-SCM-EL-10210
2000-F22-SCM-EL-10211
2000-F22-SCM-EL-10214

F22-KN-411 I-N
CARBON REGEN KILN ZONE 2
2000-F22-SCM-EL-10261
2000-F22-SCM-EL-10263
2000-F22-SCM-EL-10265

F22-KN-411 O-Q
CARBON REGEN KILN ZONE 3
2000-F22-SCM-EL-10267
2000-F22-SCM-EL-10274

F22-GE-131
CARBON TREATMENT & ELUTION EMERGENCY SUPPLY 415V GENERATOR
2000-EL019-D04-10001

TO F21-SR-001
CIL 4
415V SWITCHROOM
2000-F21-BLK-IC-10001

FROM F72-SR-001
REAGENTS
415V SWITCHROOM
2000-F72-BLK-IC-10001

ELUTION RIO PANEL RING
(F22-RIO-001 TO F22-RIO-002)
2000-F22-BLK-IC-10002

NORTHERN STAR

Approved	☒
Approved as Noted	☐
Not Approved	☐
For Information Only	

ACCEPTANCE DOES NOT RELIEVE THE SUPPLIER / CONTRACTOR OF ITS RESPONSIBILITY OR LIABILITIES UNDER THE PURCHASE ORDER / CONTRACT

NOTES

1. FOR INTERNAL FIBRE OPTIC, CAT6 PATCH LEAD (& STACK CABLING IN SWITCHROOM COMMS CABINETS) EQUIPMENT CONNECTION DETAILS REFER TO SPECIFIC TERMINATION DIAGRAMS.
2. REFER TO SPECIFIC COMMUNICATIONS TERMINATION DRAWINGS FOR EXTERNAL EQUIPMENT CONNECTION DETAIL.
3. COM CABINET, DCS PANEL, RIO PANEL, MCC & SWITCHBOARD HARDWARE/INTERCONNECTIONS (E.G. NIO NODE) & INTER-FOBOT PATCHING NOT SHOWN FOR CLARITY. REFER TO SPECIFIC COMMUNICATIONS TERMINATION DRGS FOR DETAIL.
4. ETHERNET SWITCHES LOCATED IN CABLE ZONE OF 415VAC MCC, FRONT & BACK - AS REQUIRED.
5. FIRE INDICATION PANELS SHALL BE ON A FIBRE OPTIC RING AND WILL TIE-IN TO EXISTING MAIN PLANT FIRE ANNUNCIATOR PANEL. FIP CONNECTS TO FIBRE INFRASTRUCTURE VIA 2 FIBRE OPTIC PATCH CABLES.

LEGEND

- FOB — FIBRE OPTIC BREAK OUT TRAY
- ETH — ETHERNET SWITCH HIRSCHMANN = DCS NETWORKS CISCO = ALL OTHER NETWORKS
- FCS — YOKOGAWA FIELD CONTROL UNIT
- xxxFCS4xx-NUxx — YOKOGAWA NODE INTERFACE UNIT (DUAL OPTICAL ESB PORTS: U = UPLINK, D = DOWNLINK) TAG RELATED TO PARENT FCS
- GWY — COMMUNICATIONS GATEWAY
- MC — FIBRE TO CAT6 MEDIA CONVERTER
- xxx — EQUIPMENT OR VENDOR PACKAGE PANELS AS NOTED WITHIN BLOCK.
- EOL — MCC MODULE DOL INTELLIGENT ELECTRONIC DEVICE/OVERLOAD
- UPS — UNINTERRUPTIBLE POWER SUPPLY
- VSD - VARIABLE SPEED DRIVE (WALL MOUNTED)
- MULTI CORE FIBRE OPTIC CABLE OR DUPLEX PATCH LEADS
- CAT6 — CAT6 ETHERNET CABLE CONNECTION

2000-F22-BLK-IC-10001_Rev1

F22-SR-001 415V switchroom & N-IO panels - communications block diagram sheet 1